

# No connected nontraceable detour graph has ten vertices

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## Abstract

A detour is a longest simple path. We give a computer-assisted proof that every connected nontraceable simple undirected graph on ten vertices has a vertex that is not an endpoint of any detour. A cut-vertex extension argument reduces the question to biconnected graphs. An exhaustive census contains 9,743,542 isomorphism classes: 9,707,466 have explicit Hamilton-path witnesses, and the remaining 36,076 have unequal endpoint longest-path orders, computed by exact subset recurrences. Every rejection certificate was replayed, and a separate verifier reproduced the census comparison and endpoint values. The computation depends on the documented completeness of nauty **geng** 2.9.3 and ordinary compiler and runtime semantics. Historical priority and results at other graph orders are not asserted.

## 1 Statement and context

All graphs in this article are finite, simple and undirected. Paths are simple and need not be induced; their order is their number of vertices. For a vertex  $v$  of a nonempty graph  $G$ , write

$$\tau_G(v) = \max\{|V(P)| : P \text{ is a path with endpoint } v\}, \quad \tau(G) = \max_{v \in V(G)} \tau_G(v).$$

A *detour* is a path of order  $\tau(G)$ . A graph is *traceable* if it has a Hamilton path, equivalently if  $\tau(G) = |V(G)|$ . We call a graph a *connected nontraceable detour graph*, abbreviated *CND graph*, when it is connected and nontraceable and satisfies

$$\tau_G(v) = \tau(G) \quad \text{for every } v \in V(G). \quad (1)$$

Thus each vertex must be an endpoint of some longest path; the chosen longest path may depend on the vertex.

**Theorem 1** (Computer-assisted order-ten exclusion). *Every connected nontraceable graph  $G$  with  $|V(G)| = 10$  has a vertex  $v$  such that  $\tau_G(v) < \tau(G)$ . Consequently there is no CND graph on ten vertices. In particular, there is no connected graph on ten vertices satisfying  $\tau_G(v) = \tau(G) = 9$  for all  $v$ .*

The motivating question supplied for this study was the last, explicit existence question in Theorem 1. Its source attribution is Bullock’s *Detours In Graphs*, Chapter 2, Problem 2.5.1, p. 60 [1]. That attribution, the associated historical bounds, and the current literature status have not been independently verified against the source text. They serve as provenance only. The proof below uses no external lower bound on the order of a detour.

The contribution established here is an exhaustive exclusion at exactly ten vertices, with a rejection record for every biconnected isomorphism class. This answers the specified existence question without deciding the global minimum order of a CND graph. As adjacent computational context, Zeng’s order-nine line-graph signature study [4] also uses a finite graph census and exact verification. That reference concerns a different graph invariant and supplies no detour theorem used here. Whether an equivalent order-ten exclusion already appears in the literature is undetermined; no priority claim is made.

## 2 Reduction and exact path recurrences

**Lemma 2.** *If  $x$  is a cut vertex of a connected graph, then  $x$  is not an endpoint of any longest path.*

*Proof.* Consider a path starting at  $x$ . If it has at least two vertices, all its vertices after  $x$  lie in a single component of  $G - x$ , since moving to a different component would require revisiting  $x$ . Connectivity gives a neighbor  $y$  of  $x$  in another component of  $G - x$ . Prepending  $y$  gives a longer simple path. A singleton path at  $x$  can also be extended along an incident edge. Hence every path with endpoint  $x$  is shorter than some path of  $G$ .  $\square$

At order ten, a connected graph with no cut vertex is biconnected. Lemma 2 therefore reduces (1) to biconnected graphs. An isomorphism maps paths bijectively to paths of the same order and maps their endpoints to the corresponding vertices. Both traceability and (1) are thus invariant under relabeling, so one representative of every biconnected isomorphism class suffices.

Write  $N_G(v)$  for the set of neighbors of  $v$ . For a nonempty subset  $S \subseteq V(G)$ , let  $D(S)$  be the set of possible last vertices of paths whose vertex set is exactly  $S$ . Put  $D(\emptyset) = \emptyset$ .

**Lemma 3.** *One has  $D(\{v\}) = \{v\}$  and, for  $|S| > 1$ ,*

$$D(S) = \{v \in S : D(S \setminus \{v\}) \cap N_G(v) \neq \emptyset\}. \quad (2)$$

Consequently,

$$\tau_G(v) = \max\{|S| : v \in D(S)\}, \quad G \text{ is traceable} \iff D(V(G)) \neq \emptyset. \quad (3)$$

*Proof.* Removing the last vertex  $v$  of a path on  $S$  leaves a path on  $S \setminus \{v\}$  ending at a neighbor of  $v$ . Conversely, appending  $v$  to any such path gives a simple path on  $S$ . The singleton case starts an induction on  $|S|$ . Reversing an undirected path exchanges its endpoints, which gives (3).  $\square$

The classifier represents  $S$  and  $D(S)$  by bitmasks. There are  $2^{10} = 1024$  subsets. Increasing integer mask order respects every dependency in (2), because deleting a set bit decreases the mask. All operations are exact integer operations; the endpoint orders lie between 1 and 10.

Two additional implementations check these values. A Python replay uses the forward recurrence: from each endpoint  $v \in D(S)$ , append each neighbor  $w \notin S$  and insert  $w$  into  $D(S \cup \{w\})$ . A subsequent C++ verifier instead evaluates a continuation length. For  $v \notin A$ , let  $L(A, v)$  be the maximum order of a path beginning at  $v$  whose other vertices are drawn from  $A$ . Then

$$L(A, v) = \max(\{1\} \cup \{1 + L(A \setminus \{w\}, w) : w \in A \cap N_G(v)\}). \quad (4)$$

The value 1 ends the path immediately. Otherwise its first step is to a neighbor  $w \in A$ , and deleting  $w$  from the available set prevents reuse. Conversely, prefixing  $v$  to a continuation on the right gives an allowed path. Induction on  $|A|$  proves (4), and  $\tau_G(v) = L(V(G) \setminus \{v\}, v)$ .

## 3 Exhaustive census and certificates

The enumeration uses nauty **geng** version 2.9.3. Its user guide specifies generation of all graphs of the requested class, with **-C** selecting biconnected graphs and **-g** selecting graph6 output [2, pp. 102–103]. The command is

```
geng -C -g -v 10 ten-vertices-full.g6
```

No edge interval, residue-class partition, degree parameter or additional structural restriction is supplied. The output has 9,743,542 records, with edge counts ranging from 10 to 45.

For a crosscheck of the selected domain, unrestricted generation by `geng -g -v 10` produces 12,005,168 records. A separate filter tests connectivity before and after deleting each vertex. It finds 11,716,571 connected classes and 9,743,542 biconnected classes. The retained subsequence is byte-for-byte identical to the direct `-C` output. The later verifier regenerated the unrestricted stream and repeated the subsequence comparison using a depth-first discovery/low-link articulation test. Both generations use the same `geng` software. Their agreement checks the filtering and data association; it is not an independent proof of the generator’s completeness. Completeness remains an explicit software dependency, rather than a consequence of the matching counts.

Each graph has one of two rejection certificates. An `H` record contains a permutation of  $0, \dots, 9$  giving a Hamilton path. Its validity requires only that all vertices occur once and all consecutive pairs are edges. An `N` record contains the ten endpoint orders, recomputed from the graph and checked to satisfy

$$\max_v \tau_G(v) < 10, \quad \min_v \tau_G(v) < \max_v \tau_G(v). \quad (5)$$

The classifier first searches for a Hamilton path, then runs the full endpoint recurrence if no path was found. Correctness of a rejected `H` instance relies on its explicit witness, and correctness of an `N` instance relies on the full recomputation. A failed preliminary search alone never justifies rejection.

*Exact finite observation.* The complete biconnected census has 9,707,466 valid `H` records and 36,076 valid `N` records, and no other records. Every `N` record satisfies (5). Its complete endpoint-range distribution is shown in Table 1.

Table 1: Endpoint-order ranges among all 36,076 nontraceable biconnected classes of order ten. Every range is nonconstant.

| $\min_v \tau_G(v)$ | $\max_v \tau_G(v)$ | Classes |
|--------------------|--------------------|---------|
| 4                  | 5                  | 2       |
| 5                  | 6                  | 6       |
| 5                  | 7                  | 8       |
| 6                  | 7                  | 196     |
| 6                  | 8                  | 150     |
| 6                  | 9                  | 104     |
| 7                  | 8                  | 1,937   |
| 7                  | 9                  | 1,950   |
| 8                  | 9                  | 31,723  |
| Total              |                    | 36,076  |

The Python replay directly checked all Hamilton witnesses and recomputed all endpoint vectors. The subsequent C++ verifier repeated every witness check and every endpoint computation using (4). It also checked that the graph and certificate streams ended together and matched every nontraceable vector against the supplementary graph/row/edge records. All 36 edge-stratified pairs of rejection counts agreed. For local implementation validation, the path recurrences were compared with direct path enumeration on every one of the 1,099 labeled graphs of orders one through five, including disconnected graphs. On the same graphs, the later articulation test agreed with explicit vertex-deletion connectivity tests. These small checks supplement the full certificate replay; they do not replace it.

*Proof of Theorem 1.* Let  $G$  be connected and nontraceable on ten vertices. If  $G$  has a cut vertex, Lemma 2 gives the required endpoint failure. Otherwise  $G$  is biconnected and is isomorphic to a

graph in the exhaustive census, under the stated generation dependency. That graph cannot have an  $H$  record, since  $G$  is nontraceable. Its  $N$  record therefore satisfies (5); by Lemma 3 and the replay, these are its exact endpoint orders. Some vertex has endpoint order strictly less than their maximum, which is  $\tau(G)$ . Isomorphism invariance gives the same conclusion for  $G$ . Finally, endpoint orders all equal to nine would make  $G$  nontraceable and violate this conclusion.  $\square$

## 4 Reproducibility and limitations

The accompanying computation materials retain the graph6 census, the aligned rejection certificates, the nontraceable endpoint vectors, the C++ classifier, the Python replay and the separate C++ verifier. The graph6 specification [3] fixes the upper-triangle bit order. An order-ten record begins with the byte  $I$ , followed by eight six-bit payload bytes encoding the 45 possible edges with three zero padding bits. A certificate line is  $H$  followed by ten vertex digits or  $N$  followed by ten endpoint-order digits. Each certificate refers to the same-numbered graph line. Nontraceable endpoint orders are single digits, so this format is unambiguous.

The generation used nauty 2.9.3; the retained C++ programs use C++17 and were compiled with Apple clang 16.0.0. The Python replay requires Python 3.9 or later. The reproducibility instructions specify commands and expected totals, use separate output directories, and retain input hashes in the accompanying manifest. The published numerical statement is a finite exact computation. It uses no random sampling, floating-point optimization or numerical tolerance.

The trust boundary includes **geng**'s documented exhaustive-generation contract, faithful compilation and execution, exact integer operations, and file input/output. Neither the generator nor the complete computation has been verified in a proof assistant. The different path implementations and regenerated census provide implementation crosschecks within this boundary. They do not eliminate a shared generator defect. The conclusion is confined to order ten and gives no theorem about other orders or the global minimum CND order.

**Computational and AI assistance.** OpenAI Codex assisted with the mathematical reductions, program development, execution, certificate analysis and manuscript preparation. A separate AI-assisted review context wrote and ran the continuation-based C++ verifier and checked the reduction and generation scope. This was a fresh implementation and review pass, not external peer review or formal verification. The retained programs and data, together with the arguments above, specify the evidence for the computer-assisted conclusion.

## References

- [1] Frank E. S. Bullock, *Detours In Graphs*, <https://arxiv.org/abs/2507.12086>. Bibliographic lead supplied with the motivating question; source text not inspected.
- [2] Brendan D. McKay and Adolfo Piperno, *nauty and Traces User's Guide (Version 2.9.3)*, December 31, 2025, especially pp. 102–103.
- [3] Brendan McKay, *Description of graph6, sparse6 and digraph6 encodings*, distributed format specification, updated August 2023.
- [4] Zijian Zeng, *Exact Verification of the Line-Graph Signature Inequality at Order Nine*, 2026. <https://doi.org/10.2139/ssrn.7386040>.